



GOLD TRADE ANALYSIS BETWEEN ZIMBABWE AND UGANDA

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I. Introduction

Gold is an important asset that provides stability at the time of inflation in the market and economic downfalls. It has a global demand because of its history and properties. The gold trade has been a key component in the global economy, individual wealth, and economic stability. It has got the fame as the universal standard of wealth. In the current era, gold emerges as a crucial asset. The global market of gold exports has a dynamic ecosystem that involves numerous stakeholders like miners, refineries, traders, and investing institutions.

The African region is a core source for various minerals and natural resources because of which it holds the positioning of a key player in the global gold market. Some of the major contributing countries that contribute to its significance in the market are South Africa, Uganda, Ghana, and Sudan. However, due to certain hurdles like market instability, geopolitical issues, deficit infrastructure, and economic status, the seamless export can get jeopardized leading to numerous challenges and market status. <despite all these barriers, African regions continue to stand firm as one of the leading suppliers in the gold market globally.

The African countries have a great impact on the BRICS (Brazil, Russia, India, China, and South Africa) union as majority of the gold imported in these countries are directly coming one or the other African region. Countries like China and India are the top consumers of gold due to their vast cultural diversity, beliefs, and economic factors. The increasing demand from these countries keeps the flow of African gold steady in the global market which eventually strengthens the trade relations and economic ties.

In this project, we focus on how the gold trade scenarios are in the two African countries namely Zimbabwe and Uganda. This analysis is done for the period between 2017 and 2023, which includes an analysis of the gold industry in these nations, inspecting the production, patterns in trade, influence in the market, and potential challenges faced during pre-covid, during-covid, and post-covid. For this, we have extracted the data from the website of the UN Comtrade database and extracted meaningful insights for Zimbabwe and Uganda. For this analysis, various techniques have been used such as Data Wrangling to clean the data, descriptive analysis to get statistical insights, and exploratory data analysis to get meaningful insights from the data. The whole project is implemented by using tools like Python and Excel.

II. Problem Statement

This research aims to understand the gold exports happening in Uganda and Zimbabwe for which parameters like the quantity of exportation done, the value of gold, and partner countries are taken into consideration. For a smooth flow of gold and year-on-year exponential growth, it is essential to

understand the data and extract insights for the stakeholders. These patterns are important to look for to improve future exports and global relations. The data used in this research for these two countries is taken from UN comtrade which is a global platform to get export data. However, the data comes with some challenges like missing data, unregistered countries, repetition of data, etc which is a barrier to understand the flow and scope of improvement. Additionally, the source of the dataset fails to justify the uncertainty in the data which leaves different assumptions carried out by individuals while digging into the dataset. These challenges need to be reported and worked upon for future improvements and strong global positioning in the gold export market.

III. Objective

This study aims to pull insights from the diverse gold exportation situations of Uganda and Zimbabwe. It also focuses on the different periods of exportations which are the pre- covid, during-COVID and post-COVID aspects. To understand the highs and lows in the exportation of gold different techniques were taken into action for the analysis namely Excel, EDA and Python for data cleaning and processing. The focus of the study is also to identify the geopolitical reasons, political status, economic stability, and trade terms with global partners which impact the overall market positioning of Uganda and Zimbabwe in the gold market.

IV. Methodology

Firstly, gold trade data on two African countries namely, Zimbabwe and Uganda, has been extracted through the UN Comtrade website based on a few specific parameters. The data was procured in .csv file format. Then, implementation of data wrangling techniques using Python was conducted to clean and export the data into an Excel (.xlsx) file. The following step involved application of descriptive analysis on the cleaned dataset set to compute the statistical measures like mean, median, variance, outliers and so on. Lastly, exploratory data analysis (EDA) was performed on the data to discover further insights into the gold trade by Zimbabwe and Uganda.

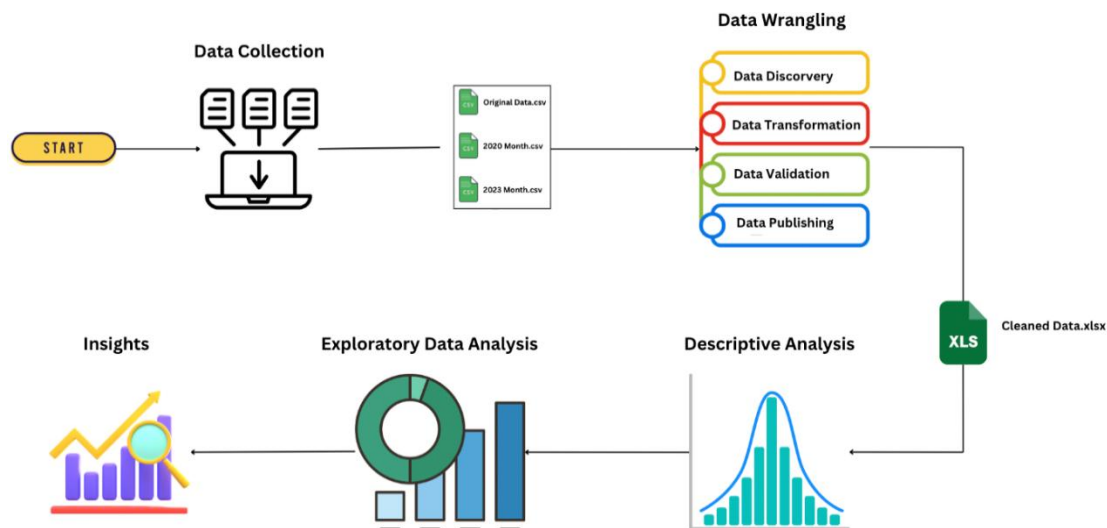


Figure 1 WorkFlow PipeLine

A. Data Collection

The Gold-Trade data on mainly two African nations, Zimbabwe and Uganda has been obtained. The first data file (Original Data.csv) contains valuable information on the yearly trading of Gold by these two fore-mentioned countries throughout the period 2017 to 2023. This data comprises of 117 rows and 47 columns. Some of the notable columns in the data are listed below alongside their description.

Column Name	Description
Period	Years in which Gold is traded by Zimbabwe and Uganda
ReporterDesc	Name of the designated African Countries which traded Gold
PartnerDesc	Name of the partner countries to whom the Gold is traded with
Qty	The total quantity of Gold traded to the countries
PrimrayValue	The total value (in dollars) of Gold traded to the countries

Table 1 Description of Some Attributes in the Dataset

It turned out that the attribute 'Qty' displayed multiple zero (or null) values in 2020 and 2023. To tackle this anomaly, monthly data was also obtained through the UN Comtrade database for these two years. The data files titled '2020 Month.csv' and '2023 Month.csv' hold information on the monthly gold trade data by Zimbabwe and Uganda in 2020 and 2023 respectively. The month-wise data for 2020 consists of 156 rows and 47 columns whereas the monthly data for 2023 has a total of 56 rows and 47 columns.

B. Data Wrangling

a. Data Discovery

In this step, all the datasets are loaded into Python in form of a data frame. Moving ahead, the analysis of the datasets is done to check the missing values, duplicate rows and potential outliers. The datasets were thoroughly examined to check for the presence of null values, or duplicate data. Some missing

values were detected; however, no duplicate data was found. Subsequently, the columns of the dataset are classified into categorical (or qualitative) variables and numerical (or quantitative) variables. The outliers are checked for, using boxplots on the numerical variables.

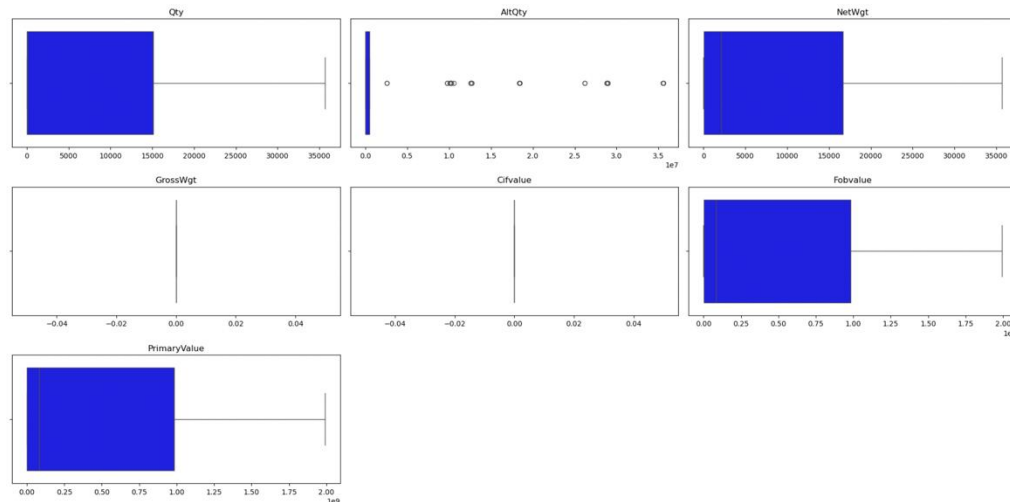


Figure 2 Boxplot to check the outliers before Data Transformation

b. Data Transformation

In this step, necessary columns are extracted from the datasets to help in analyzing the gold trade data. The data for the years 2020 and 2023 in the original dataset (Original Data.csv) is replaced by the monthly data extracted earlier, for these two years. A new column labelled as 'PricePerQuantity' is introduced to the data by dividing the values in the column 'Primary Value' by the values in the column 'Qty'. The column 'Primary Value' is then checked for DIV error, and data expressing this error are ejected.

Lastly, it is detected that the total quantity and the total primary value in 'PartnerDesc' as 'World' is equal to the total quantity and the total primary value for the rest of the countries respectively. Therefore, the data containing 'World' as 'PartnerDesc' is also removed.

Period	Reporter	World	Other Countries	Total Qty
2017	Zimbabwe	43800.046	43800.046	87600.092
2017	Uganda	21166.66	21166.66	42333.32
2018	Zimbabwe	56222	56222	112444
2018	Uganda	25576	25576	51152
2019	Zimbabwe	47160.36	47160.36	94320.72
2019	Uganda	57565.062	57565.062	115130.124
2020	Zimbabwe	31580.38	35037.38	66617.76
2020	Uganda	54814	65466	120280
2021	Zimbabwe	45147.076	45147.076	90294.152
2021	Uganda	36888.48	36888.48	73776.96
2022	Zimbabwe	71412.356	71412.356	142824.712
2022	Uganda	0	0	0
2023	Zimbabwe	37366.8	57887.72	95254.52
2023	Uganda	0	0	0

Table 2 Proof that the quantity of World is equal to the total quantity of importers

Period	Reporter	World	Other Countries	Total Primary Value
2017	Zimbabwe	1772051148	1772051148	3544102296
2017	Uganda	831341484.7	831341484.7	1662682969
2018	Zimbabwe	2283057994	2283057993	4566115987
2018	Uganda	1031128054	1031128054	2062256108
2019	Zimbabwe	2127828020	2127828020	4255656040
2019	Uganda	2513501176	2513501176	5027002352
2020	Zimbabwe	1964515436	1964515386	3929030822
2020	Uganda	3635957842	3635908988	7271866830
2021	Zimbabwe	3225686712	3225686712	6451373424
2021	Uganda	2065709114	2065709114	4131418228
2022	Zimbabwe	3984693266	3984693266	7969386532
2022	Uganda	0	0	0
2023	Zimbabwe	3610953836	3585838390	7196792226
2023	Uganda	0	0	0

Table 3 Proof that the total value generated by World is equal to the total value generated by importers

c. Data Validation

This step is initiated by thorough re-inspection of the transformed data. This is performed by replicating the initial steps conducted in Data Discovery. The transformed data has 76 rows and 8 columns of data. Similar to data discovery, null values and duplicate data are checked for in the dataset. It is concluded that there are no missing values and no duplicate data present. Next, the columns of the dataset are re-categorized into categorical (or qualitative) variables and numerical (or quantitative) variables. The outliers are checked for, using boxplots on the numerical variables for the transformed data.

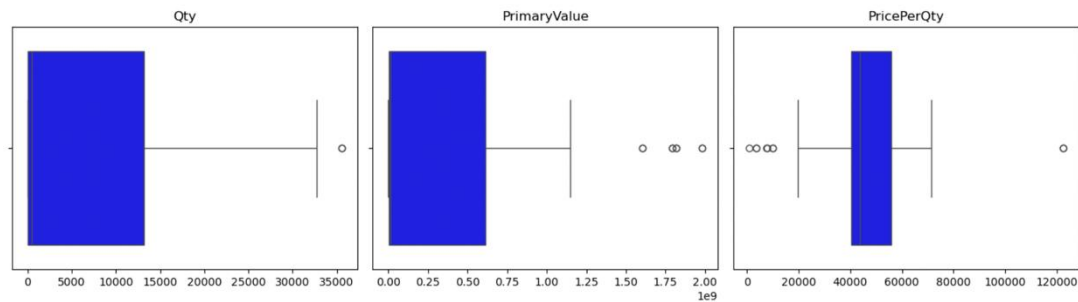


Figure 3 Boxplot to check the outliers after Data Transformation

d. Data Publishing

This is the final step in Data Wrangling. This is an indication that the data has been cleaned and is free from null values and duplicate data. The data is published in an Excel file for further analysis.

C. Descriptive Analysis

This analysis gives a deeper insight of the various parameters of gold export scenarios in Uganda and Zimbabwe. It tells about various key factors like max gold sold, value at which the price is sold, average price of gold, deviation etc. It is an essential step taken for in depth research of the gold trades happening within and outside the African continent.

Zimbabwe and Uganda recorded the highest quantity of gold trade with the United Arab Emirates (UAE) in 2022 (35531 kg) and 2019 (32733 kg) respectively. Uganda recorded its second-highest gold trade of 2.6 tons with Turkey in 2019. Zimbabwe recorded its second-most quantity of gold trade at 17164 kg with South Africa in 2017. This shows that the quantity of gold traded between our reporting countries and UAE from 2017 to 2023 is significantly greater than trade quantities with other partner countries, as indicated by the outlier.

	count	mean	std	min	25%	50%	75%	max
ReporterDesc								
Uganda	33.0	6262.490970	10344.332937	0.001	2.0000	87.82	9765.33	32733.00
Zimbabwe	43.0	8294.579953	10476.683320	0.123	118.0295	2128.17	14301.20	35531.07

Figure 4 Statistical Analysis of Quantity

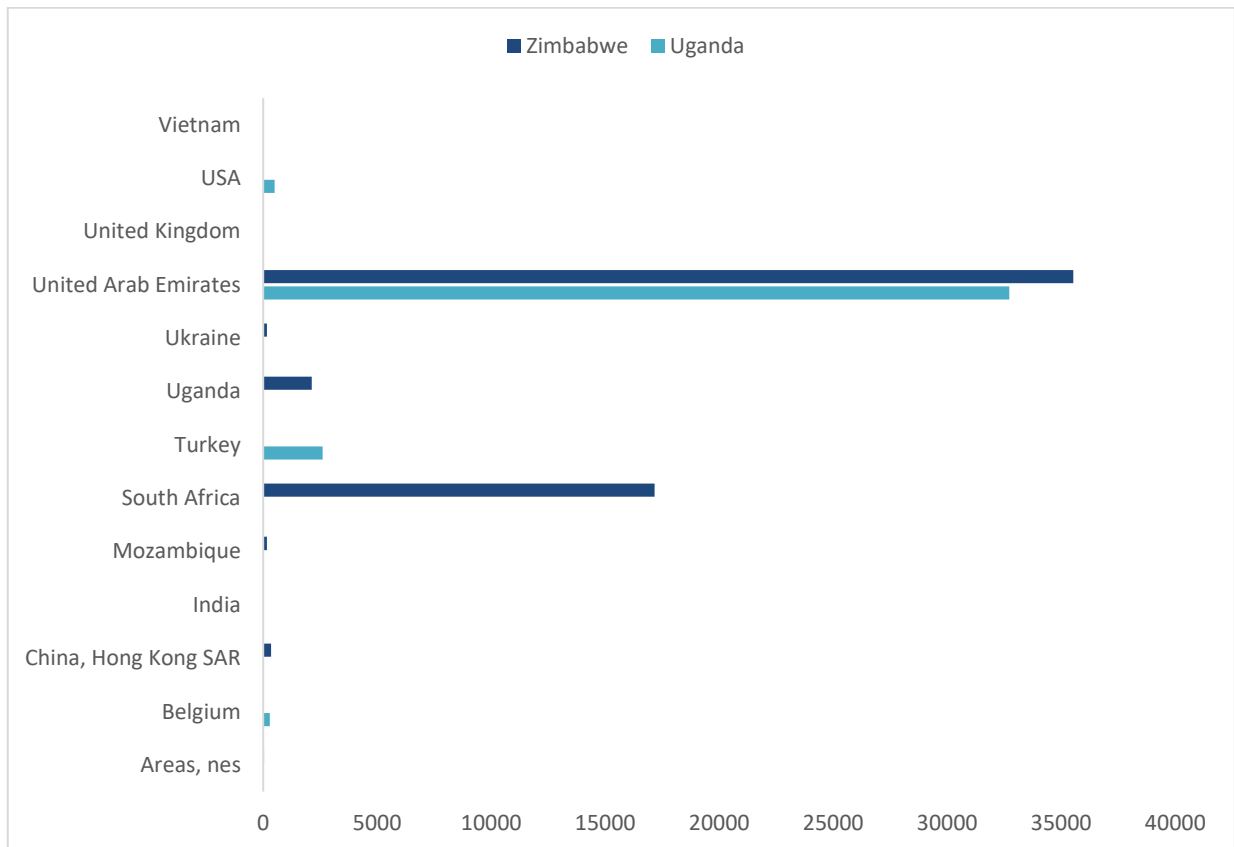


Figure 5 Bar Graph to show maximum Quantity exported to the countries

Zimbabwe boasts of a consistently high value of gold traded with UAE which increases over the period and records its peak in 2022 with a massive trade of 1.98 billion dollars. Uganda recorded its highest value of gold trade with the UAE in 2020 with around 1.82 billion dollars' worth of gold traded. Apart from UAE, Zimbabwe had significantly high values of trade with South Africa at 6.92 billion dollars and Uganda at 1.27 billion dollars in 2017 and 2020 respectively.

	count	mean	std	min	25%	50%	75%	max
ReporterDesc								
Uganda	33.0	3.053815e+08	5.351726e+08	9.97	82897.500	1.786869e+06	400380291.4	1.817954e+09
Zimbabwe	43.0	4.405505e+08	6.120714e+08	100.00	6040422.255	1.271939e+08	644777010.1	1.981811e+09

Figure 6 Statistical Analysis of Trade Value

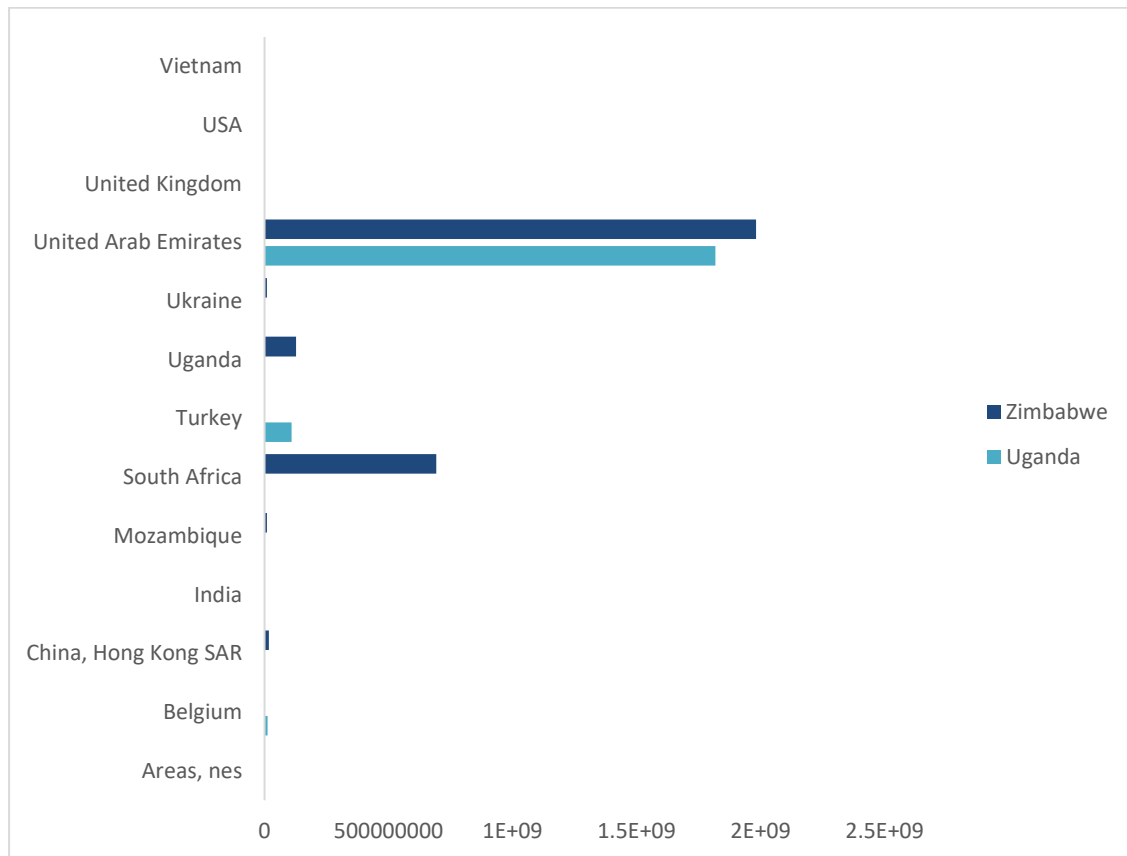


Figure 7 Bar Graph to show maximum Trade Value

Uganda recorded a surprisingly huge trade of gold by price per kg with Mozambique in 2020 coming to 122451 dollars per kg of gold indicated by the single extreme outlier. This is because the quantity of gold traded to Mozambique is much less compared to the total primary value Uganda received by exporting the gold to this country. This shows that there might be a discrepancy in data regarding Mozambique. Zimbabwe marks a low trade in gold by price per kg, with South Africa at a mere 813 dollars in 2017 and Uganda showed a very low trade in gold by price per kg with USA, China & Hong Kong SAR, and UAE in 2017, 2018, and 2019 respectively shown.

	count	mean	std	min	25%	50%	75%	max
ReporterDesc								
Uganda	33.0	34448.690265	16211.985204	3573.73870	22408.563165	41448.750000	42500.761229	56005.53621
Zimbabwe	43.0	53945.762374	19236.917390	813.00813	43032.327332	55167.115714	59766.790251	122451.59518

Figure 8 Statistical Analysis of Price per Quantity

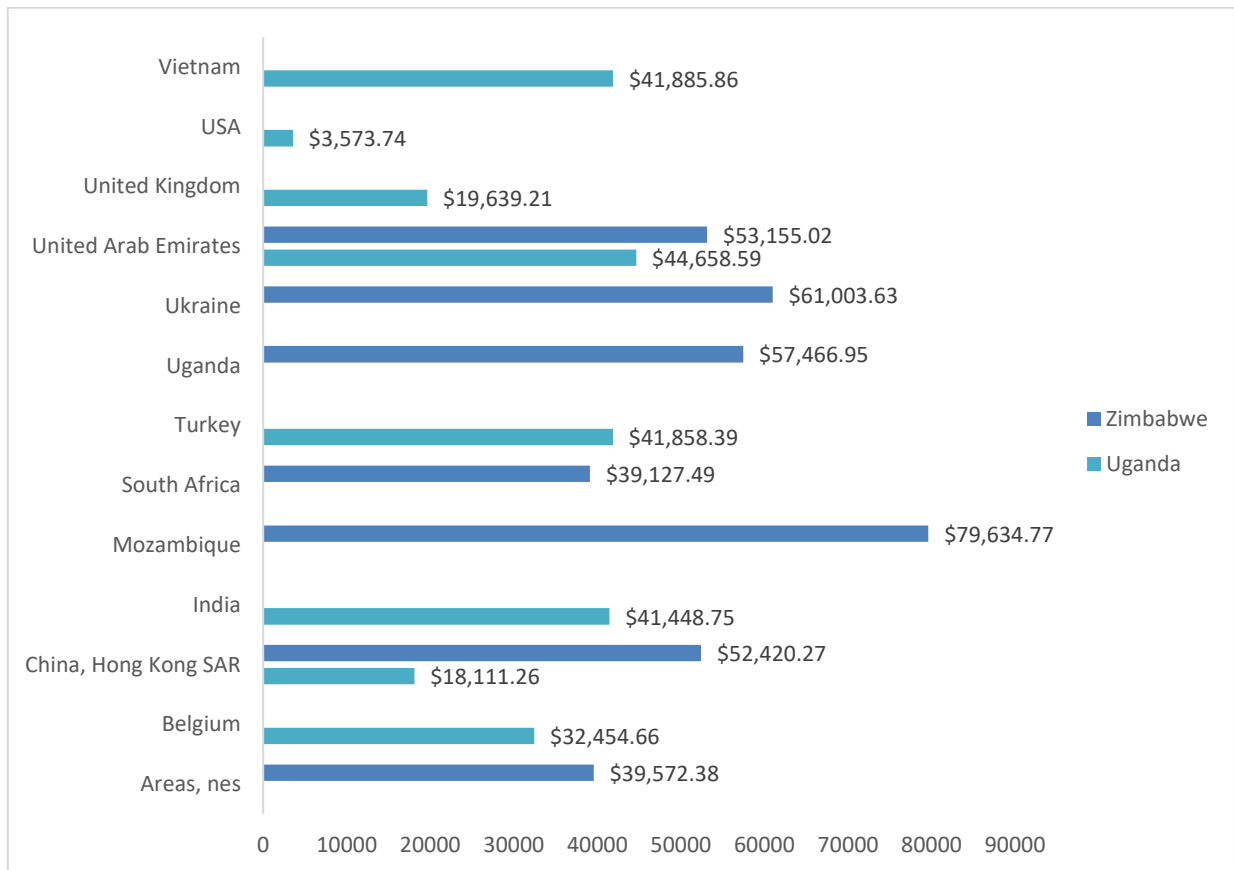


Figure 9 Bar Graph to show the maximum Price per Quantity

D. Exploratory Data Analysis

1) Gold Trade Analysis on Uganda:

Uganda is an East African country known as the “Pearl of Africa”. Natural resources like gold, copper and oil are the attraction of Uganda. Uganda’s gold deposit stand as a pillar for the country’s economy. Major regions are Karamoja, Busia, Baweja and Kigezi for gold mining. Gold trade is particularly significant in Uganda [11]. Uganda has become a local hub for illegal gold trade from other parts of East Africa because of its attractive market.

In 2019, gold trade revenue in Uganda rose up to US\$2.5 billion than the previous year. Kampala’s refiners act as an intermediate for smuggling the gold between the mines and international illegal exporting [12].

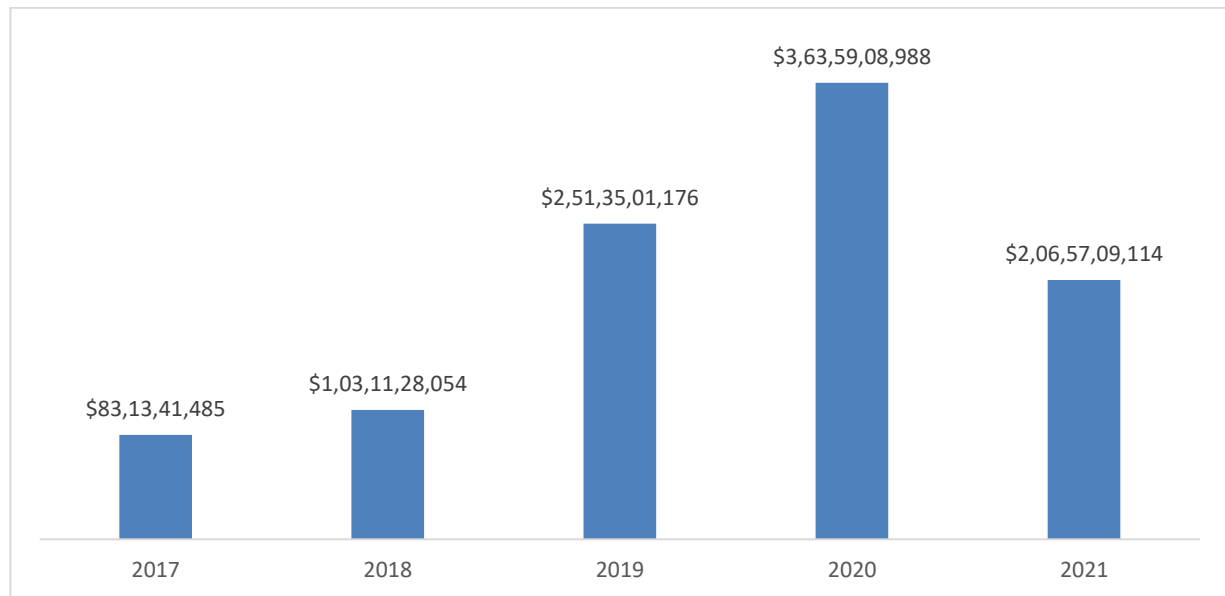


Figure 10 Total Trade Value of Gold Exported by Uganda in each year

Price of the gold various places in Uganda indicates the smuggling route, lowest prices at mining sites and local region and highest prices in Kampala and main exporting centers [13]. The graph shows average price per quantity of exported gold by Uganda from the year 2017 to 2021. The average price per quantity of exported gold from Uganda shows a moderate decrease from 2017 to 2018, constant in 2019, and then there was a gradual increase in 2020, stabilizing that higher level through 2021. This suggests a notable change in market conditions or export strategies starting in 2020.

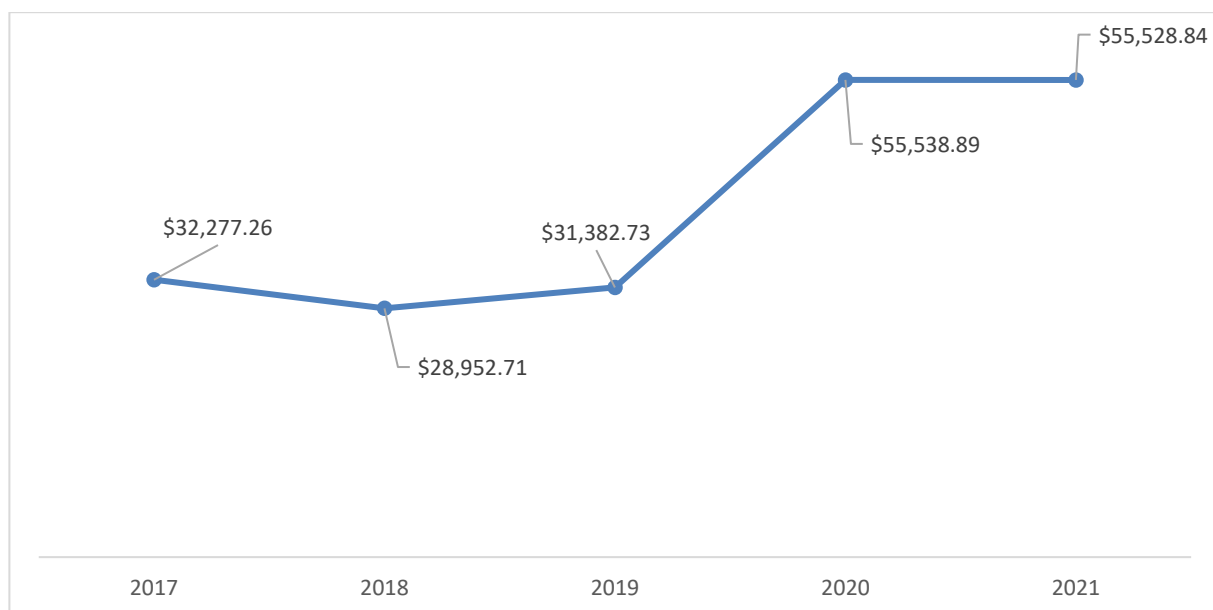


Figure 11 Average Price per Quantity of Gold exported by Uganda in each year

After 2021, a lot of data regarding Uganda's gold trade were missing. The tax issues, which started in July 2021, led to disagreement over taxes imposed on gold exports and refiners. "From the information provided by the Uganda Revenue Authority (URA), only two companies are willing to pay the tax. The

other companies have raised concerns, and this is being evaluated,” the minister told parliament. The government of Uganda and exporters/dealers of gold had a difference of opinion on tax which led to the suspension of gold shipments for the majority of 2021's second half. The government's initiative to levy higher taxes on gold exports led to this conflict which lasted for more than 2 years. Due to this issue, the export of gold in Uganda reportedly dropped to Zero for a few months in late 2021[1][2].

The Bank of Uganda found that, previous year, \$ 2.2 billion worth of gold was exported and no gold exporting during 2021-2022, which indicates the decreasing official exports denotes, a huge amount of gold might have shifted or transported through illegal routes without being noticed. Researchers assume that certain exports might have operated illegally in violation of the Law or through trafficking so that they can avoid recently implemented taxes and regulations.[3]

Before 2021, Uganda's gold trade was marked by a lack of transparency with limited information about the origins and amounts of gold being exported [4]. Investigation on Uganda's gold trade reported that significant difference between gold production and exporting figures, major portion of gold were exported or shifted illegally without proper documentation. However, a latest updating of Uganda's gold trade is significant recovery of gold trade. In the last year up to February 2024, gold trade generated \$2.7 billion, almost 37% of overall profit from exporting. This is an important part for controlling the tax disputes but the complete details are remains to be sorted.

Uganda's Gold Trade Analysis Pre-COVID-19, During COVID-19 and After COVID-19:

Before the COVID-19 pandemic, there was a major difference between reported gold production and export figures in Uganda's gold trade including a lack of transparency in the industry. The country emerged as an important player in the regional gold trade mainly as a hub for re-export of gold brought from nearby countries. The trade was marked by minimal oversight resulting in trafficking and inaccurate reporting. Regardless of these issues, Uganda's economy began to depend primarily on the gold industry which also boosted revenues from exports.

The COVID-19 pandemic and following lockdowns had varied impacts on Uganda's gold trade. In smuggling hubs like Arua, buyers have been purchasing Congolese gold and sold it local market rather than transporting it to Kampala. However, Through Entebbe airport huge amount of gold were masked and transported as jewelries worn by passengers on flight to UAE and India, hence lockdowns had slight impact on gold smuggling in Uganda. The pandemic even started new ways smuggling, gold was transported as covid-19 aid in the cargo planes from Uganda to UAE and other markets. Although the pandemic, Uganda's gold trade remained strong, earning over US\$120 million in March and April 2020, according to the central bank.

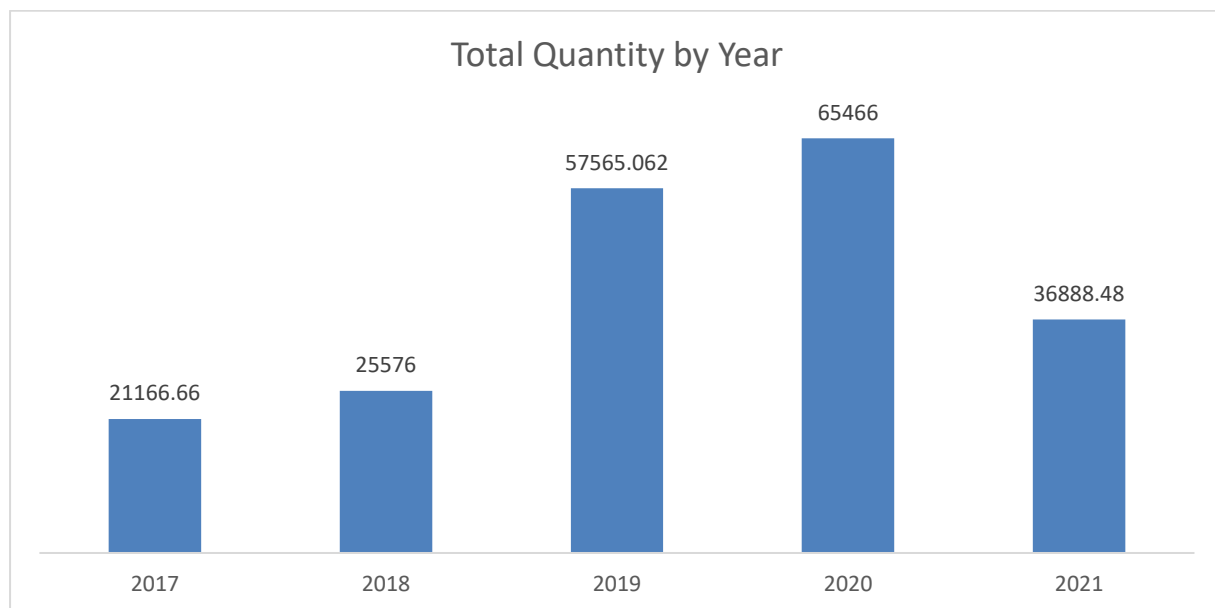


Figure 12 Total Quantity of Gold Exported by Uganda in each year

The COVID-19 pandemic led a collapse global trade and supply chains, which might have a chance of severe impact on Uganda's gold trade with countries like China, Turkey, and Vietnam. However, new trade route to the UAE initiated more flexible and stable during this breakdown. The UAE, particularly Dubai, has been a major destination for Uganda's gold exports for years. The search results indicate that the UAE typically purchases around 98% of Uganda's exported gold [5]. This existing strong trade relationship likely made it easier to maintain exports during the pandemic.

After Covid-19 pandemic, global demand and high price was fixed for the gold during the pandemic which led a positive impact on Uganda's gold trade. The country kept a significant export figure, through the trade of gold earned good revenue. However, a lot of obstacles are there, even though work is under processing for improving the regulatory control and transparency. Changes in the dynamics of international trade such as the Shipment of Russian gold trade to Hong Kong and intensified inspection in the UAE, have also influenced Uganda's gold trade recovery. These changes have provided Uganda with new opportunities in the global gold market.

Uganda's Gold Exports to the UAE:

Due to several factors, Uganda's gold exports to UAE have shown a substantial spike before, during, and after the COVID-19 pandemic. As a major regional hub, Uganda has become an entry point for illegal gold smuggling from nations nearby such as South Sudan and the Democratic Republic of Congo (DRC), which is then subsequently sent to trade hubs like UAE [7]. Uganda's gold exports to the UAE went up from \$300 million in 2018 to almost \$2 billion by the end of 2022, implying an active and expanding trade relationship [8][9]. During lockdown, Uganda's gold trade flourished despite the pandemic with gold being illicitly transported and then sold locally in hubs like Arua. This did not affect trade through Entebbe airport, where gold was usually smuggled in bulk or as jewelry worn by travelers. Cargo planes

delivering COVID-19 aid became a popular method for smuggling gold to Dubai and other overseas market. Despite the pandemic Uganda made nearly \$120 million in March and April 2020, by exporting gold [7]. Dubai as a financial center in the Middle East along with UAE's regulatory and enforcement loopholes has allowed for the trade in conflict and high-rise gold, making it an increasingly important destination for Uganda's gold export [8][10]. After the covid, Uganda has noticed a rise in international travelers and overall trade, which has boosted the exporting of gold to the UAE [8].

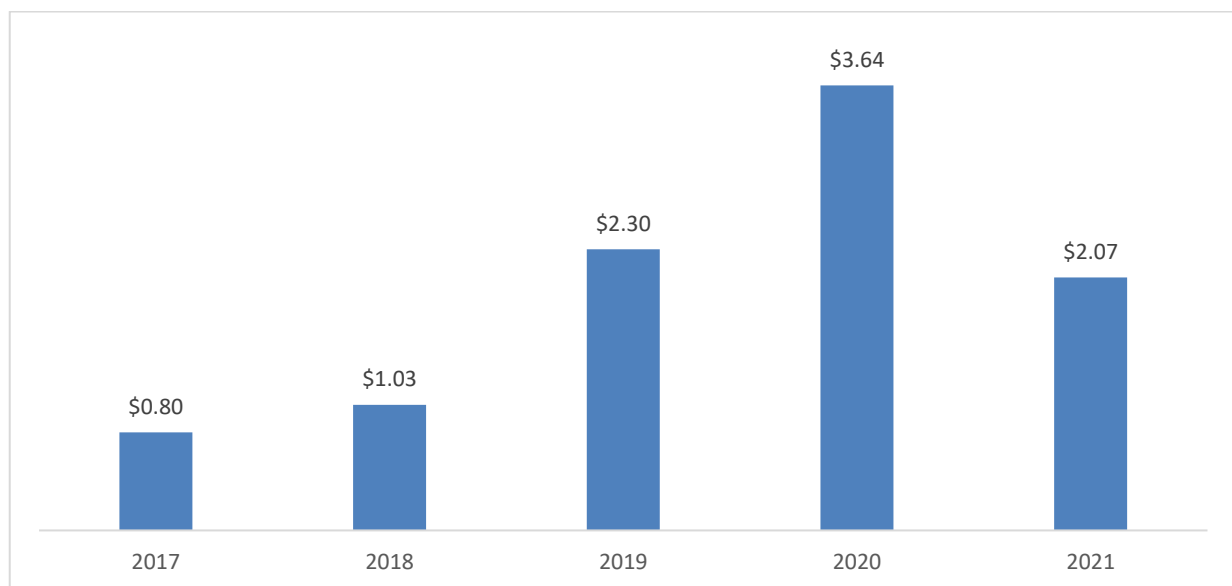


Figure 13 Total Trade Value of Gold Exported to the UAE by Uganda

2) Gold Trade Analysis on Zimbabwe:

Zimbabwe is a key player in the gold trade, which is a major component of its economy [14]. The gold sector is supported by both large-scale mining companies and small-scale miners. However, illegal mining activities, economic instability, and the COVID-19 pandemic have significantly impacted this sector. This portion of our analysis will concentrate on the gold trade in Zimbabwe, analyzing events leading up to, through, and including COVID-19 as well as evaluating the impacts of illicit mining operations.

Gold exports are one of Zimbabwe's primary sources of income [15]. The country produced 45 tons of gold in 2021, increased production to 71 tons in 2022, and decreased to 58 tons in 2023. Gold exports significantly contribute to Zimbabwe's economy, generating substantial annual export revenues.

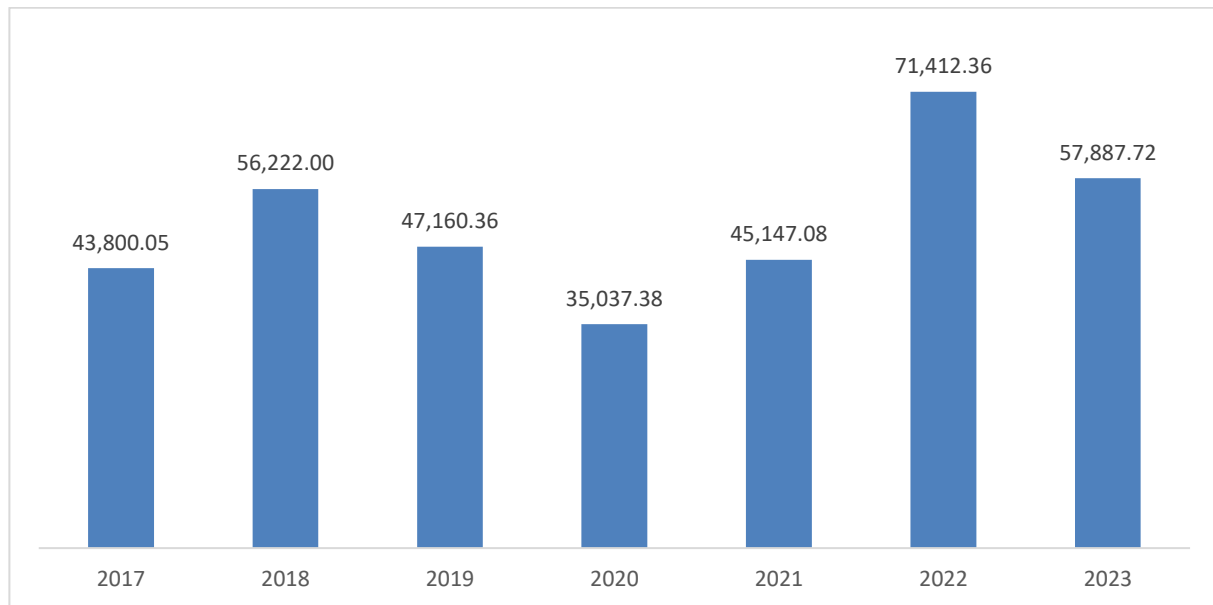


Figure 14 Total Quantity of Gold exported by Zimbabwe in each year

The United Arab Emirates receives a hefty share of gold exports—roughly 75% of the entire amount from 2017 to 2023 [16]. Only 23% of the total quantity of gold is exported to South Africa. A very small portion of gold is exported to the countries like China, Mozambique, Uganda, and Ukraine.

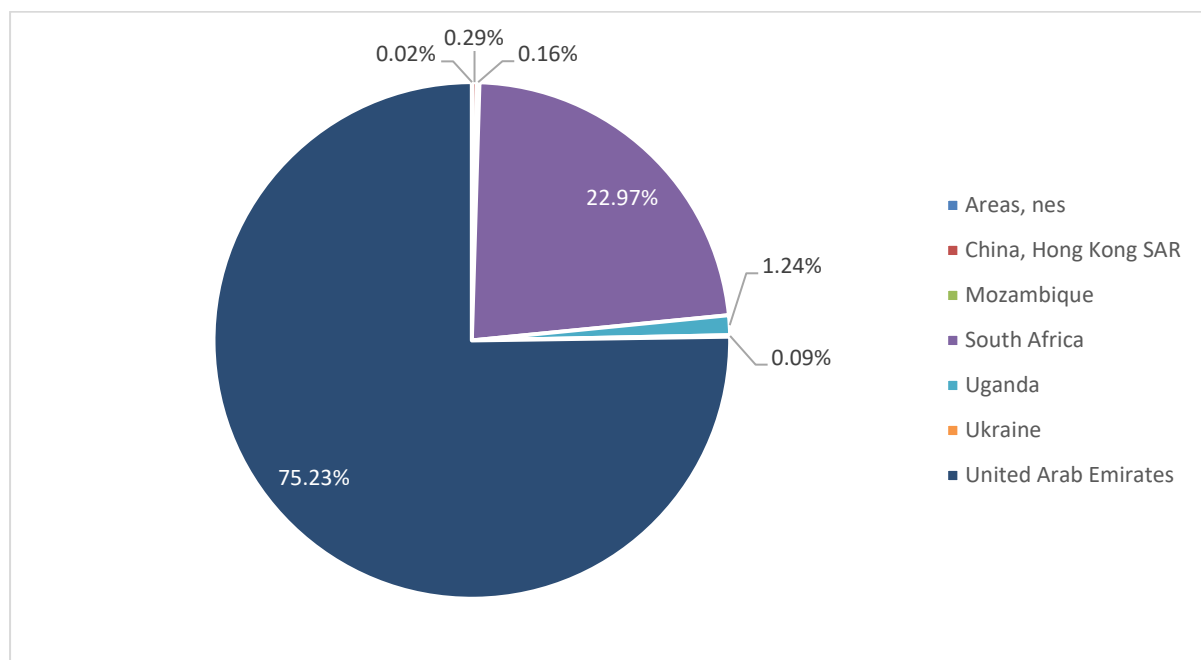


Figure 15 Distribution of the countries to which the Gold is exported by Zimbabwe

Gold's price per kilogram varies in response to shifts in global markets. Certain highs and lows were experienced in the entire time frame because of different possible factors. Between 20017 and 2019 a steep hike in the price per kilogram of gold can be seen from \$32679 dollars in 2017 to around \$66561 dollars by the year 2020. In the COVID phase of year 2020 a drop in the price of per kg of gold was seen which kept on falling until 2022. In 2022 an average price of \$58129 was experienced which by the end

of 2023 reached \$61944.

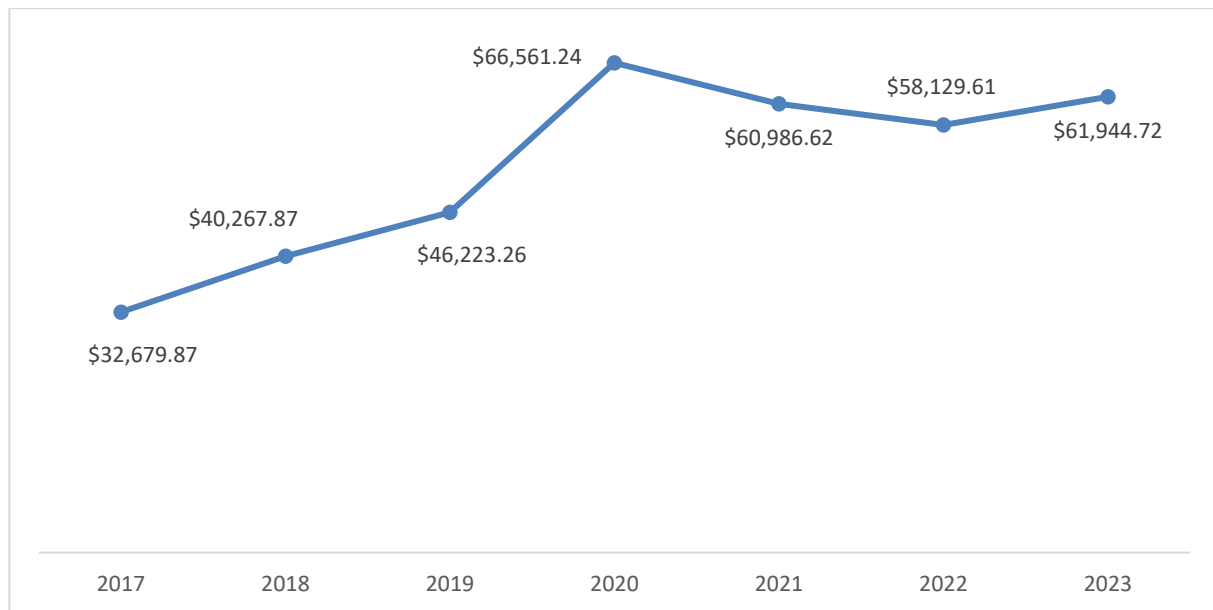


Figure 16 Average Price per Quantity of the Gold exported by Zimbabwe in each year

Zimbabwe's Gold Trade Analysis Pre-COVID-19, During COVID-19 and After COVID-19:

Before the COVID-19 pandemic, Zimbabwe's gold trade was very active, supported by both large and small-scale mining companies [17]. However, illegal mining activities were widespread, causing significant economic and environmental problems. Illegal mining is hurting Zimbabwe's economy a lot. About \$100 million worth of gold is being smuggled out of the country each month. This reduces the country's official gold export money by a lot.

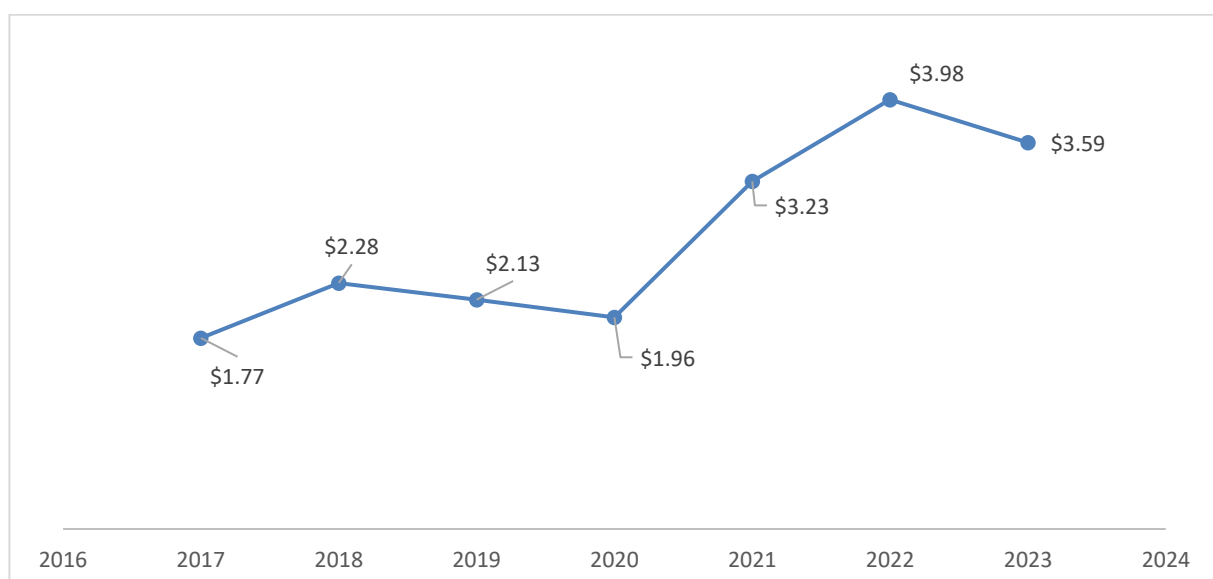


Figure 17 Total Trade Value of Gold exported by Zimbabwe in each year (in billions)

During the COVID-19 pandemic, Zimbabwe's gold trade was badly affected [18]. At the start of the pandemic, gold prices went up and down because of global problems, hurting Zimbabwe's income. Restrictions, transport issues, and not enough workers made mining hard, reducing gold production and exports. Many people turned to illegal gold mining because of economic difficulties, making the problem worse. COVID-19 made Zimbabwe's weak economy even weaker, causing more revenue loss and making it harder to stop illegal trade.

After the pandemic, Zimbabwe's gold trade is trying to improve. But recovery is hard. High inflation, unemployment, and economic problems still hurt the gold trade. Illegal mining adds to the hardship. The government is making new rules to stop illegal mining and control the gold trade. These rules aim to make the gold trade better and more profitable. But, bigger and long-term plans are needed to fully stop illegal activities.

Reasons for the Decline in Gold Trade in Zimbabwe in 2023:

In 2023, there was a noticeable decline in Zimbabwe's gold trade. The main reasons for this are economic uncertainties and fluctuations in global market conditions. High inflation and unemployment rates have caused a contraction in domestic demand, and global economic stagnation and uncertainties have negatively affected gold demand. Additionally, the government's new taxes and rules have made it hard for gold exporters to operate, stopping trade in some cases.

Gold Trade Between Zimbabwe and the UAE Before BRICS Membership:

The UAE joined the BRICS group in 2024. Before this, the UAE imported a lot of gold from Zimbabwe. The UAE joined BRICS to increase economic cooperation and expand global trade networks. This is part of the UAE's plan to grow its economic and political influence. Trade data before the UAE joined BRICS shows that much of Zimbabwe's gold exports went to the UAE. This trade helped both countries and set the stage for future cooperation.

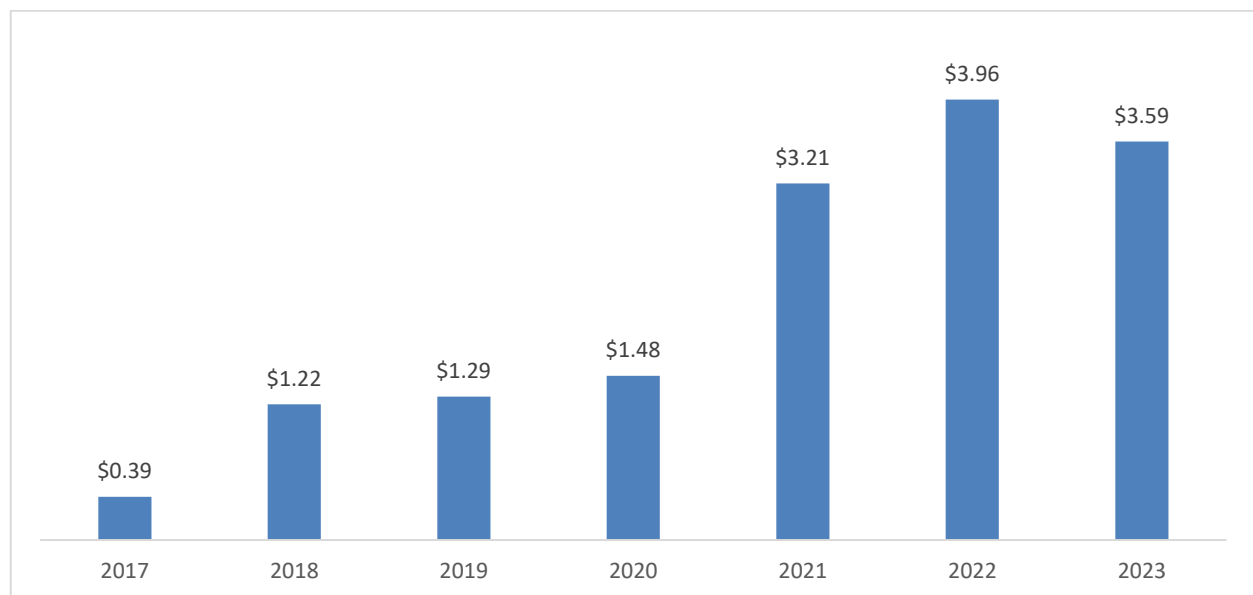


Figure 18 Total Trade Value of Gold exported to the UAE by Zimbabwe

3) Analysis of BRICS Nations

The BRIC countries are a grouping of foreign investment strategies among rising economic powers. This group initially comprised four countries Brazil, Russia, India, and China which were at a similar stage of economic advancement during the early 21st century. In the initial times the main goal of BRICS was development and cooperation on international affairs and in a more wider view financing for development, economic cooperation, social and cultural exchanges, technology and innovation, sustainable development, peace, and security. In the August 2023 summit a decision was taken to open doors for new members into this organization. The new BRICS+ organization now consists of four more countries: Egypt, Ethiopia, Iran, and the United Arab Emirates, it now accounts for 37.3% of world GDP (Saudi Arabia is also expected to join the BRICS). By the end of 2022, it had provided about \$32 billion to emerging nations for infrastructure development projects. China wants to grow its power and influence and become the voice of the global south through BRICS. "Russia sees [BRICS] as part of its fight against the West, helping it to overcome the sanctions imposed after the invasion of Ukraine," says Creon Butler of the London-based think tank Chatham House. Russia is holding the presidency of 2024 BRICS, they look to increase the role of BRICS in the international financial system, develop cooperation between banks, expand BRICS currency usage, and promote collaboration between tax and customs authorities [19][20].

Zimbabwe actively participates in gold trade with BRICS nations primarily with China, India, Russia, and South Africa. To fight hyperinflation currency woes to several other challenges, the Central Bank made a strategic decision to stockpile gold to leverage the valuable mineral assets of Zimbabwe. It has also mandated that mining companies pay a portion of their royalties in gold, this pushes Zimbabwe towards adopting a currency backed by gold. BRICS will be the largest buyer of gold in 2023. China needs

Zimbabwe's mineral resources particularly gold and diamonds to empower its manufacturing and technological industries. India's jewelry market has a huge demand for gold. South Africa is a destination for mining and refining as well as transit point for gold smuggling. South Africa holds a pivotal role with trading relationship with Zimbabwe in trade of mining equipment, food products, manufactured goods and live and work in South Africa policy which helps to bolster their foreign exchange. Russia has on deal with Zimbabwe for major mining project like Darwendale Platinum Project also some major investments in gold mining. In addition to that Russia cooperate with Zimbabwe in military and technology exchange. Depending on the international market rates the pricing of gold traded between Zimbabwe and the BRICS nations vary [21]. During 2017 the gold price in Zimbabwe peaked at \$43,375 per Kg, in 2018 it peaked at \$43,608, in 2019 it peaked at \$49,879. Subsequently during Covid i.e., in 2020 gold price peaked at \$66,529, in 2021 it peaked at \$61,348, in 2022 it peaked at \$65,716, in 2023 it peaked at \$66,620 and currently they are trading around \$75K.

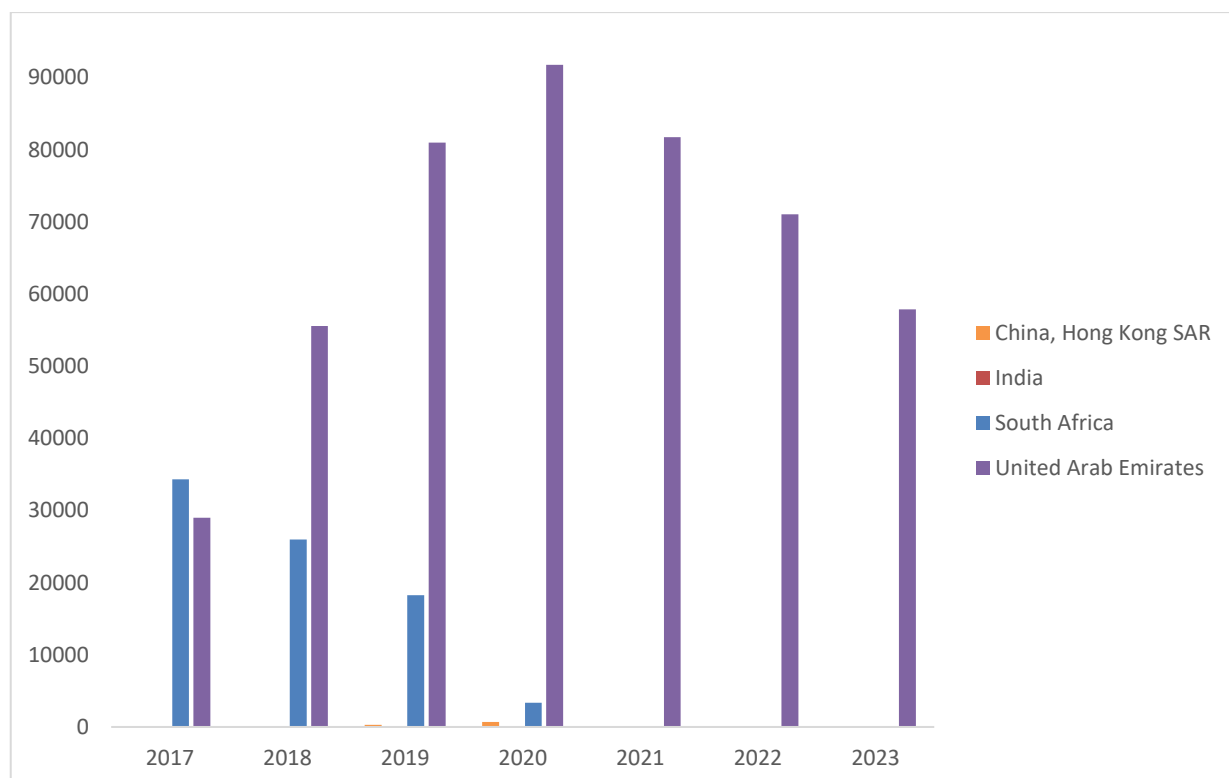


Figure 19 Total Quantity of the Gold exported to the BRICS Nations by Uganda and Zimbabwe

Uganda has traded gold with BRICS nations like China and India during 2017-2021 and exported substantially to UAE and middle east. In addition to that trade with USA and Belgium seems much higher potential than the BRICS countries. Gold prices are fluctuated with international demand and supply statistics while in the case of Uganda in the period analyzed above the price-maintained relation with international price and the average price of gold trade was 1254\$ in 2017 was increased to 1900 in the 2024. Uganda's gold trade peaked in the year 2020 to 65466 unit of gold which is the highest all over the period from 2017 to 2023. Despite various challenges Uganda maintained a high gold export and in 2017 it peaked at \$120 million in June. Uganda sold 21166 Kg of gold in 2017 but trading with BRICS

nations accounted for nil. While on the subsequent year 2018 Uganda traded 17 unit of gold where major portion of 13 unit accounted for China and Hong Kong and 4 unit was traded with India. Furthermore in 2019 Uganda gold trade with BRICS nation declined to 2 unit particularly with China. Subsequent years 2020 and 2021 Uganda haven't traded with BRICS countries major trade was with the UAE.

During 2017-2023 UAE was a prominent partner in gold trade with Zimbabwe though UAE was not part of BRICS then. By analyzing the data of gold trade, it is visible that the average quantity of gold traded to be 10,044.19 Kg per year and the standard deviation to be 11,432.78 Kg for Zimbabwe. The average trade value was found to be \$515.31 million / year with a standard deviation of \$647.02 million. Furthermore, Zimbabwe gold trade data is vastly higher than the Uganda gold trade showing its larger production capabilities. In addition to that Zimbabwe gold trade revenue also much higher than Uganda correlating the quantity of trade. It is recorded that the peak period of gold trade of Zimbabwe was in 2018 which accounted about 56222 unit.

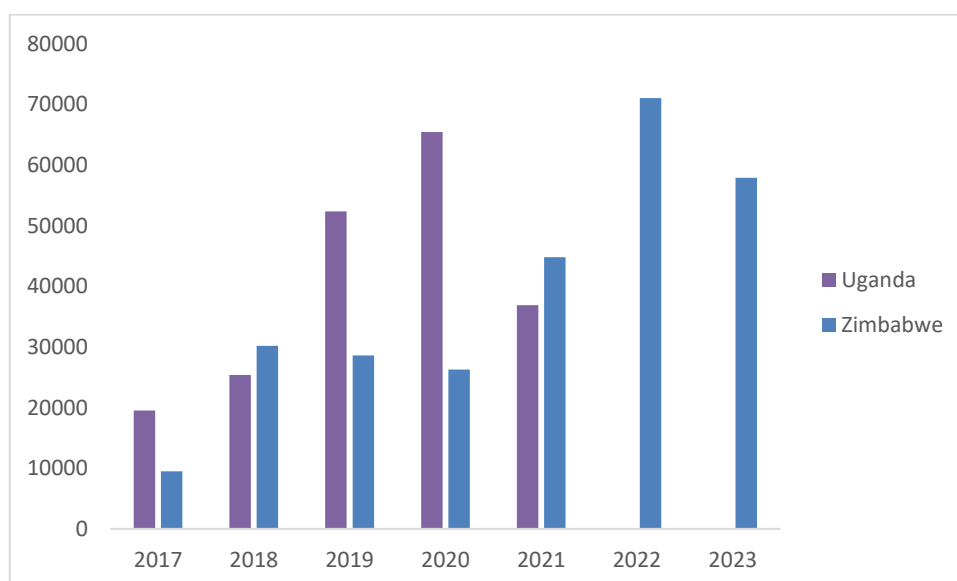


Figure 20 Total Quantity of Gold exported to the UAE by Uganda and Zimbabwe

Due to the established network with BRICS nations, Zimbabwe traded higher volumes of gold with BRICS compared to the USA and Europe. It had a higher trade value and price variability, but stable with European countries. Uganda had partnered with Belgium, USA and other European countries, it had low trade volume with USA and Europe compared to BRICS which had moderate volume. Uganda had higher price per kg with USA and Europe with premium pricing for certified gold with a more stable price.

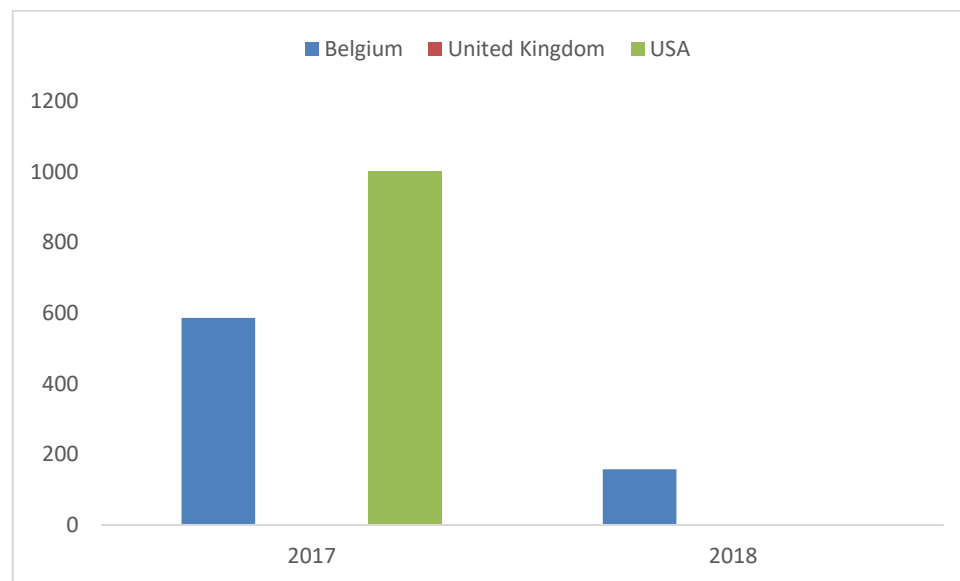


Figure 21 Total Quantity of Gold exported to Belgium, USA and UK by Uganda

V. Results

Figures 22 and 23 demonstrate a comparison between Zimbabwe and Uganda based on the total quantity of gold the total trade value of gold exported respectively.

As compared to Zimbabwe which actively traded before the pandemic, Uganda showed a moderate increase in trade figures from 300 million USD in 2017 through 2018 with a sudden spike in 2019 and further increase in 2020, making nearly 120 million USD in just March and April of 2020, owing to its ever-expanding trade relationship with UAE who typically purchases around 98% of Uganda's total gold exports.

The advent of Covid-19 in mid-2020 severely hit Zimbabwe's gold market hurting the economy. Restrictions, transportation problems, and inadequate manpower resulted in a steep fall in gold production and exports. This encouraged illegal gold mining, further weakening the economy and stacking losses.

On the other hand, Uganda's trade with UAE flourished throughout the pandemic, displaying revenue worth 2 billion USD generated by the end of 2022, owing to the discovery of a new trade route. Trade with other countries like China, Turkey and Vietnam fell during this period.

Post-pandemic figures for gold trade in Uganda were missing or unreported due to internal factors like tax disputes leading to illegal exports and smuggling to evade taxes and regulations. Zimbabwe, on the contrary, managed to stabilize and increase production from 45 tons in 2021 to 71 tons of gold in 2022, generating solid revenue in this period. This shift in the scale can be attributed to the government's constant efforts to prevent illegal mining and control the gold trade. A noticeable decline in Zimbabwe's gold trade in 2023 can be observed mainly due to economic uncertainties and fluctuations in global

trade market in addition to the new tax regulations imposed which made it difficult for gold exporters to operate, halting trade in some cases.

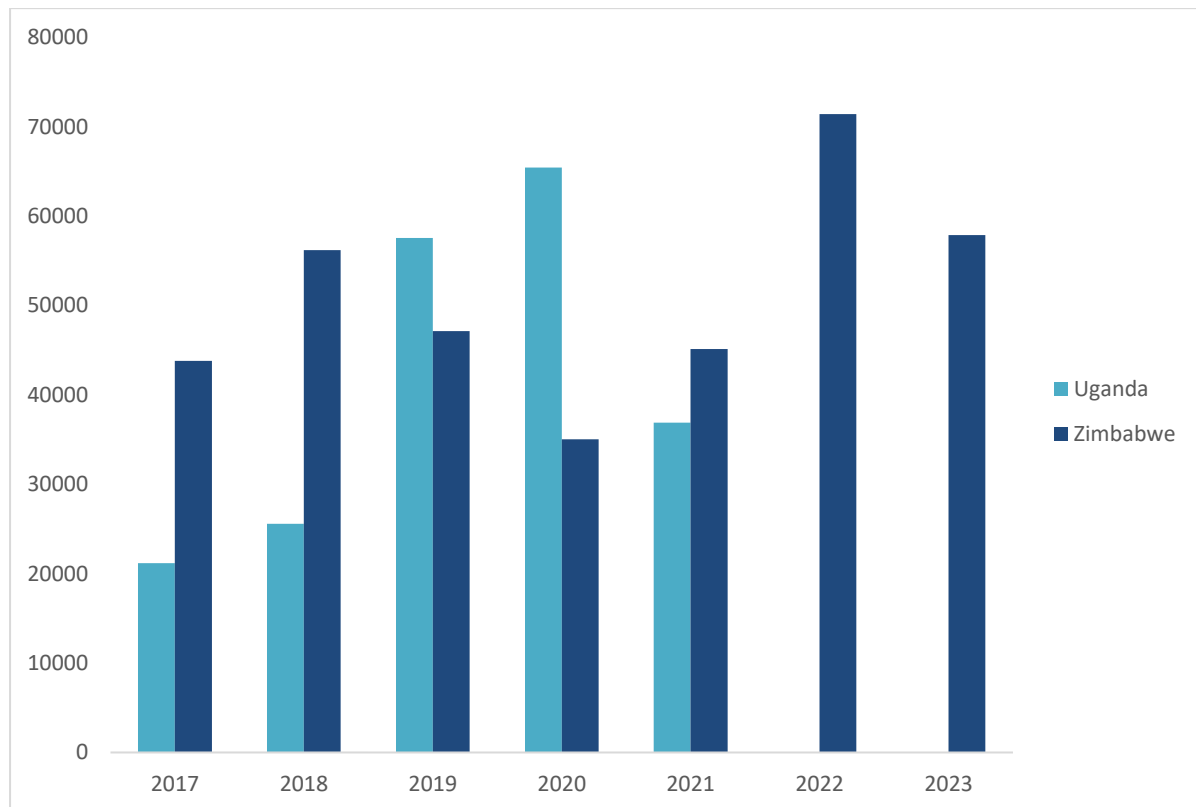


Figure 22 Comparison on total quantity of Gold exported by Uganda and Zimbabwe

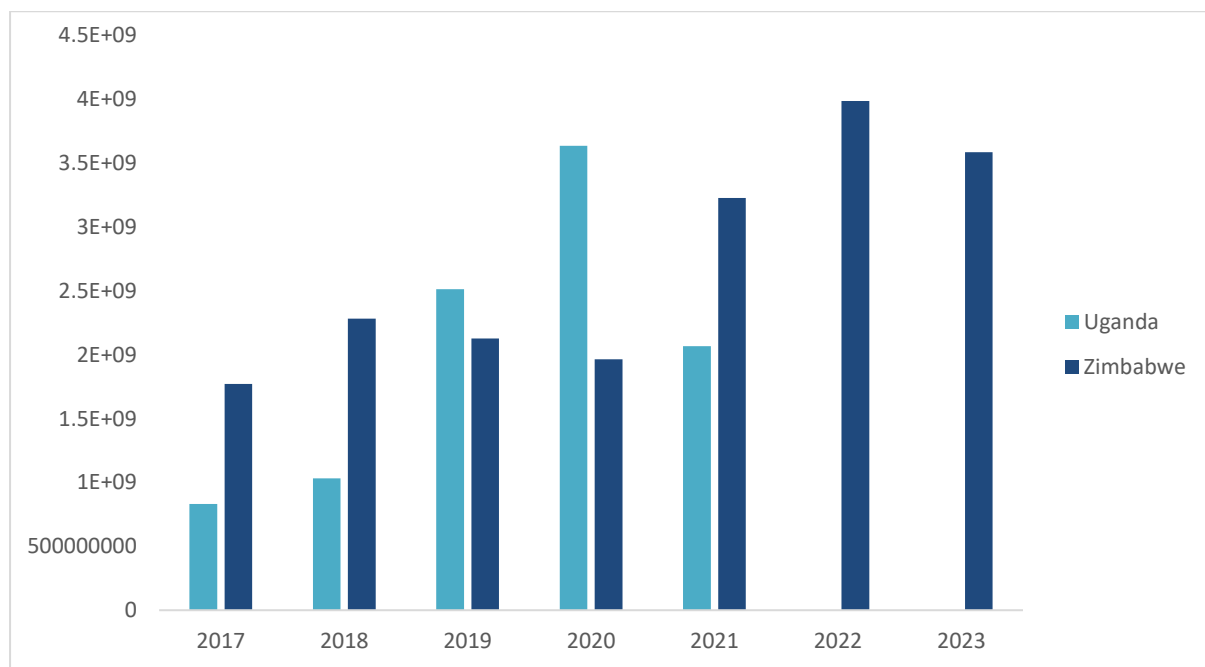


Figure 23 Comparison of the total Trade value of the Gold exported by Uganda and Zimbabwe

Zimbabwe boasts of a consistently high value of gold traded with UAE which increases over the period and records its peak in 2022 with a massive trade of 1.98 billion dollars. Uganda recorded its highest value of gold trade with the UAE in 2020 with around 1.82 billion dollars' worth of gold traded. Uganda recorded a surprisingly huge trade of gold by price per kg with Mozambique in 2020 coming to

122451 dollars per kg of gold and, to China and Hong Kong SAR in 2020 valued at 58393 dollars per kg.

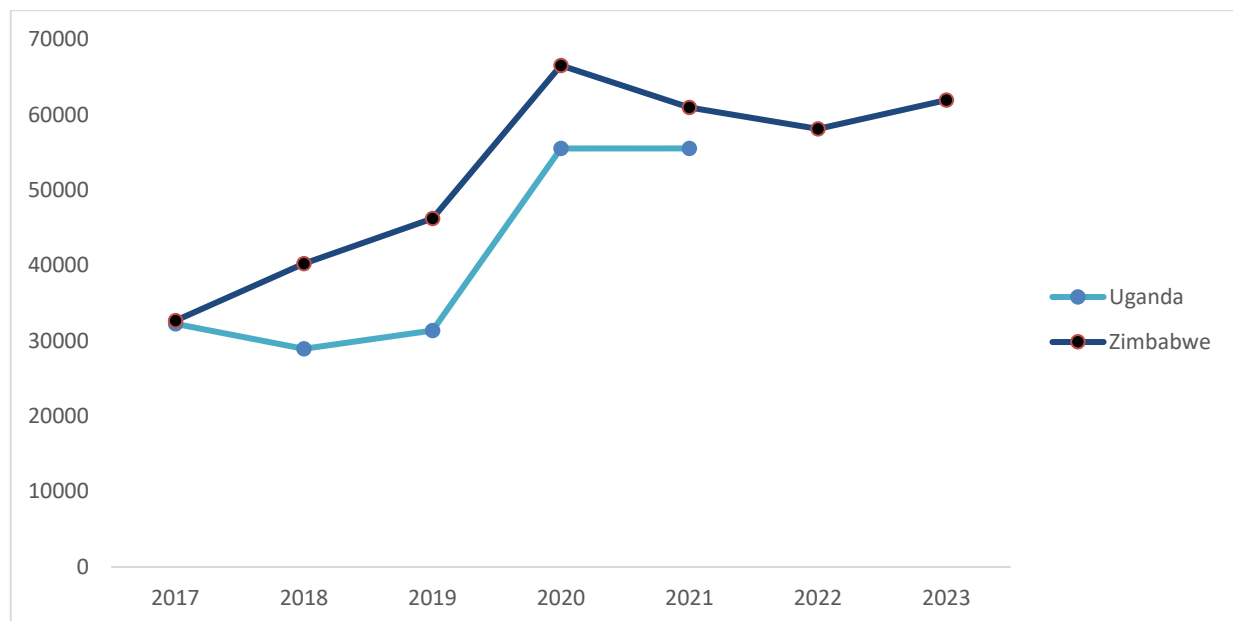


Figure 24 Comparison on average price per qty at which the Gold is exported by Zimbabwe and Uganda

Conclusion

Zimbabwe's gold trade is very important for the country's economy. Gold exports make up a big part of Zimbabwe's income and help its economic growth. During the COVID-19 pandemic, Zimbabwe's gold trade was severely affected. The pandemic caused disruptions in global trade, logistical challenges, and labor shortages, negatively impacting gold production and exports. Economic uncertainties and price fluctuations during the pandemic led to a decrease in Zimbabwe's gold revenues. The pressure on the healthcare system increased, there were disruptions in basic services, and poverty rates rose.

New government rules and economic uncertainty caused the drop in gold trade in 2023 [6]. High inflation and jobless rates led to lower domestic demand. Global economic problems also reduced demand for gold.

Zimbabwe's gold exports benefit from trade with BRICS nations, which also gives economic stability. Global market conditions impact trade prices with BRICS countries. The UAE joined BRICS to expand global trade and improve economic cooperation. Before joining BRICS, Zimbabwe's gold exports to the UAE were good for both countries. The United Arab Emirates receives a hefty share of Zimbabwe's gold exports—roughly 75% of the entire amount from 2017 to 2023 [16]. Only 23% of the total quantity of gold is exported to South Africa. A very small portion of gold is exported to the countries like China, Mozambique, Uganda, and Ukraine.

Illegal mining was prevalent in Zimbabwe even before Covid-19 and posed a threat to the nation's

economy [7]. These activities increased during the pandemic, causing more revenue loss for the government. After the pandemic, the government is making new rules to fight illegal mining and control the gold trade. But bigger and long-term plans are needed to fully stop these activities.

Despite the serious impacts of the COVID-19 pandemic and economic problems on Zimbabwe's gold trade, government actions and international cooperation can revive the sector [8]. Making big plans for a sustainable and regulated gold trade and fighting illegal mining is essential. This way, Zimbabwe's gold trade can greatly help the country's economic growth and stability.

Uganda, also known as the "Pearl of Africa", is a storehouse of natural resources like gold, copper and oil which contribute largely to the country's economy. The gold trade revenue of Uganda had a drastic rise to 2.5 billion US dollars in 2019 with the country's capital, Kampala acting as an intermediate for gold smuggling and illegal export as reported by an investigation showing a significant difference between gold production and exporting figures and stated that there was no gold exported in 2021-2022 compared to 2.2 billion USD worth of gold traded in the previous year.

However, a lot of data on Uganda's gold trade post 2021 was found to be missing. This anomaly can be attributed to rising tax issues in July 2021 leading to a conflict between the government and gold export traders which resulted in trade figures dropping to zero in late 2021.

Uganda's economy solely relied on the gold industry before the onset of the pandemic and remained strong even during covid-19 due to the introduction of a new trade route to UAE who purchased 98% of Uganda's exported gold during that period. The country managed to follow up with a steady recovery post the pandemic and currently stands with myriads of opportunities in the global gold trade market.

Zimbabwe actively trades gold with BRICS nations, particularly China, India, Russia, and South Africa, to combat economic challenges. The Central Bank mandates mining royalties in gold, positioning Zimbabwe toward a gold-backed currency. Notably, South Africa serves as a transit point for gold, and Russia is involved in major mining projects in Zimbabwe.

Uganda's gold trade with BRICS is minimal compared to its significant exports to UAE and other global markets. Despite facing challenges, Uganda maintained high gold exports, peaking in 2020. While its trade volume with BRICS is lower, Uganda's gold pricing aligns closely with international rates, ensuring competitive market presence.

Both countries benefit from their gold trade relationships with BRICS, leveraging these interactions to bolster their economies amidst varying market conditions.

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